

Project 3

SQL Data Analysis

DecodeLabs Data Analytics Internship

Prepared By

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Tools Used

Microsoft SQL Server

Microsoft Excel

Microsoft Word

GitHub

Project Objective

To perform SQL-based data analysis on an e-commerce dataset using SQL queries for filtering, sorting, grouping, and aggregation in order to generate meaningful business insights.

Submission Date

June 2026

Project Overview

This project was completed as part of the DecodeLabs Data Analytics Internship Project 3.

The objective of this project is to use SQL queries to extract, analyze, and summarize business information from an e-commerce dataset. SQL Server was used to perform data retrieval, filtering, sorting, grouping, and aggregation operations.

The project demonstrates the practical use of SQL fundamentals including SELECT, WHERE, ORDER BY, GROUP BY, COUNT(), SUM(), and AVG() functions to generate business insights from transactional data.

Dataset Overview

The dataset contains e-commerce transaction records including Order ID, Customer ID, Product, Quantity, Unit Price, Payment Method, Order Status, Referral Source, Coupon Code, Items in Cart, and Total Price.

The dataset was analyzed to understand customer purchasing behavior, product performance, revenue generation, payment preferences, and order processing trends.

Data Preparation

Before importing the dataset into SQL Server, the data was reviewed to ensure consistency and accuracy.

The dataset was prepared with standardized date formats, validated records, consistent data types, and complete business information. The cleaned dataset was then used as the source for SQL analysis.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemInCart	CouponCode	ReferralSource	TotalPrice
1	ORD200000	04/01/2023	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903		7 SAVE10	Instagram	2853.1
2	ORD200001	23/08/2024	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779		3 SAVE10	Referral	302.7
4	ORD200002	27/02/2024	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982		8 FREESHIP	Email	2753.4
5	ORD200003	15/10/2023	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070		5 SAVE10	Facebook	273.19
6	ORD200004	08/05/2025	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424		8 SAVE10	Email	2504.04
7	ORD200005	23/10/2023	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927		4 SAVE10	Instagram	491.72
8	ORD200006	17/06/2025	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362		6 SAVE10	Facebook	664.42
9	ORD200007	12/05/2023	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193		9 FREESHIP	Facebook	747.75
10	ORD200008	02/04/2025	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692		2 No Coupon	Email	268.56
11	ORD200009	21/11/2023	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363		6 SAVE10	Google	2037.52
12	ORD200010	29/12/2023	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248		9 WINTER15	Instagram	3129.85
13	ORD200011	17/02/2024	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646		7 SAVE10	Email	147.42
14	ORD200012	15/10/2024	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991		6 FREESHIP	Referral	361
15	ORD200013	30/08/2023	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874		4 FREESHIP	Facebook	604.47
16	ORD200014	27/03/2023	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539		3 SAVE10	Instagram	786.66
17	ORD200015	17/07/2023	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938		3 No Coupon	Google	473.96
18	ORD200016	24/01/2023	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259		3 FREESHIP	Email	1067.62
19	ORD200017	02/03/2024	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116		3 No Coupon	Referral	423.4
20	ORD200018	26/03/2025	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045		3 SAVE10	Facebook	431.44
21	ORD200019	23/01/2023	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365		9 SAVE10	Google	1120
22	ORD200020	08/05/2023	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651		9 WINTER15	Instagram	312.55
23	ORD200021	17/11/2024	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983		6 No Coupon	Google	1371.8
24	ORD200022	29/01/2023	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350		2 SAVE10	Google	99.24
25	ORD200023	09/04/2025	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083		5 FREESHIP	Referral	1093
26	ORD200024	30/10/2024	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057		3 WINTER15	Referral	461.9
27	ORD200025	24/07/2024	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327		3 WINTER15	Email	399.98
28	ORD200026	06/03/2024	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812		6 No Coupon	Referral	297.75

Working in SQL

Database Creation

To perform SQL-based analysis, a dedicated database named ECommerceAnalysis was created in Microsoft SQL Server. Creating a separate database helps organize project data and provides a structured environment for executing SQL queries and storing analytical results.

The database serves as the central repository for all e-commerce transaction records used throughout this project.

The following SQL command was used to create the database:

```
CREATE DATABASE ECommerceAnalysis;
```

After creation, the database was selected as the active working environment using:

```
USE ECommerceAnalysis;
```

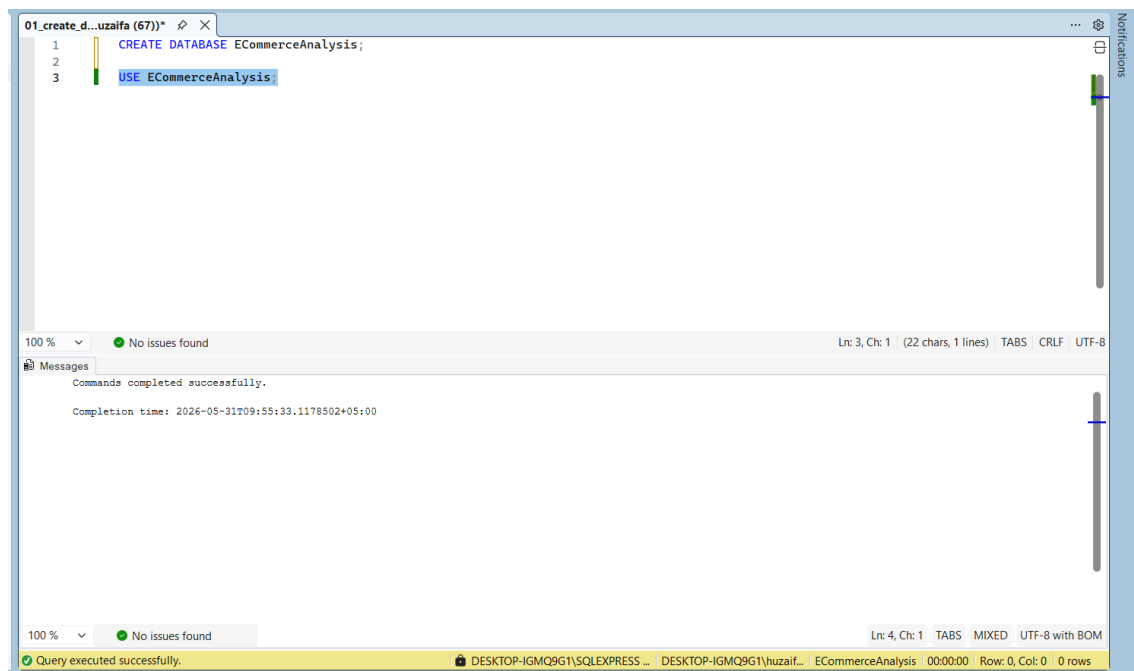


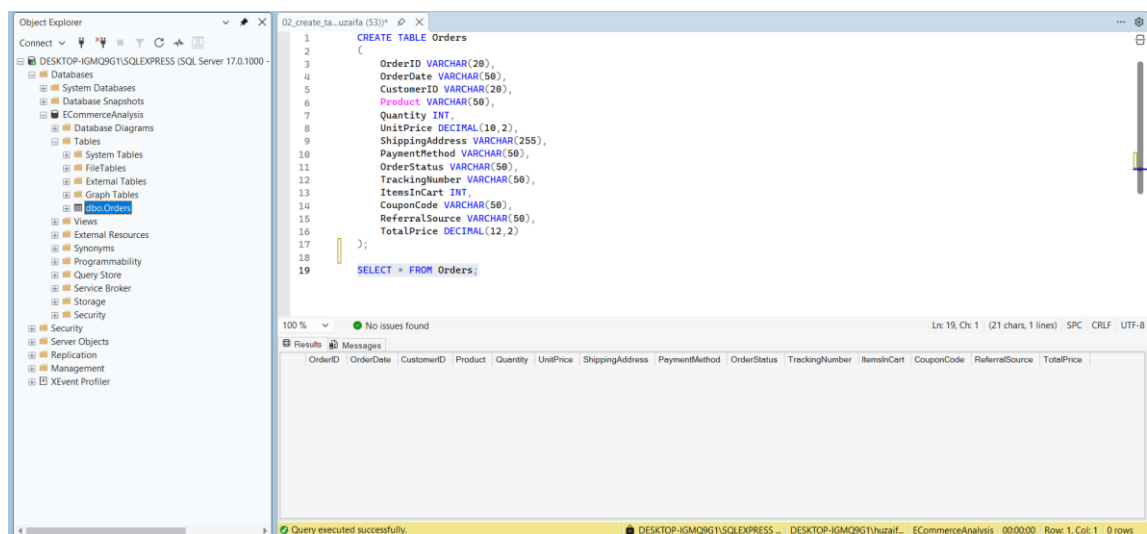
Table Creation

After creating the database, a table named **Orders** was created to store the e-commerce transaction records. The table structure was designed based on the dataset columns and included fields related to customer information, product details, payment methods, order status, referral sources, coupon codes, and revenue.

Appropriate SQL data types such as **VARCHAR**, **INT**, and **DECIMAL** were assigned to ensure accurate storage and analysis of data. During the initial implementation, the **OrderDate** column was defined using the **DATE** data type. However, while importing data from Excel, SQL Server generated date conversion errors because the source file contained date values in a format that was not directly compatible with the DATE field during the import process.

To ensure successful data loading, the **OrderDate** column was temporarily defined as **VARCHAR(50)** during the import stage. This approach allowed all records to be imported successfully without data loss. Date conversion and validation can be performed later during data analysis if required.

The Orders table serves as the primary data source for all SQL queries and analytical operations performed during this project.

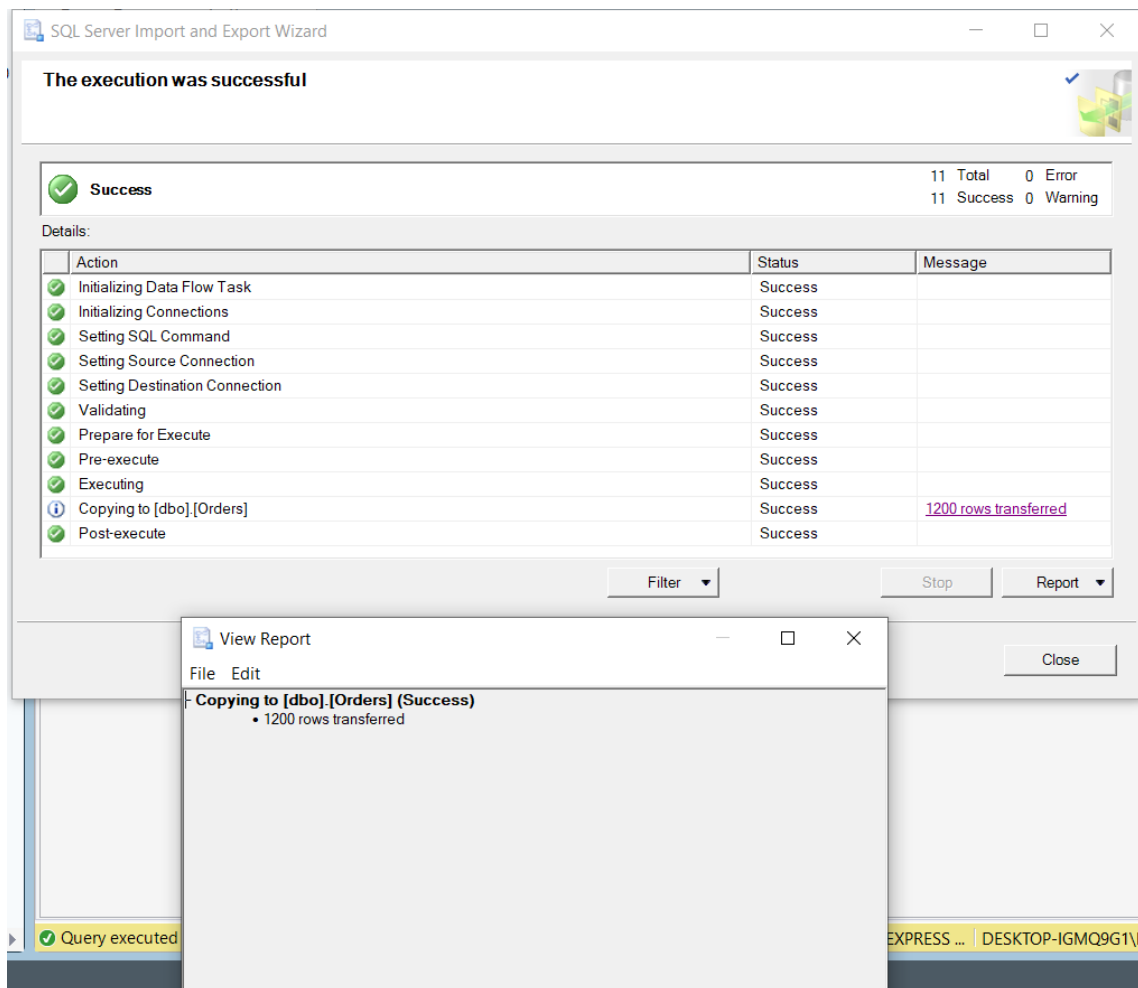
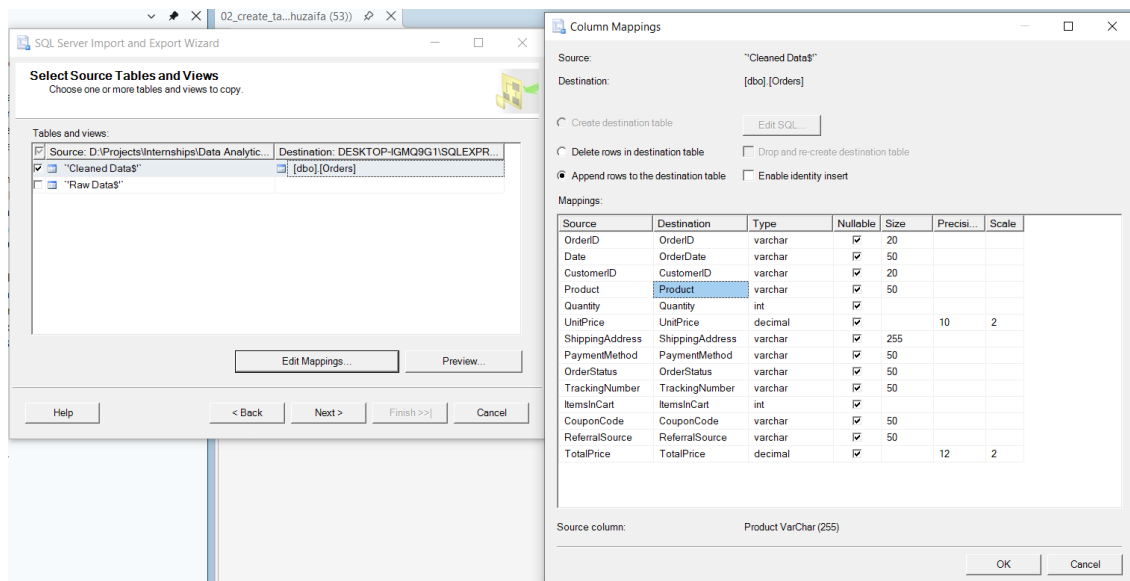


Data Import Process

After creating the Orders table, the cleaned dataset was imported into SQL Server using the **SQL Server Import and Export Wizard**. During the import process, column mappings and data type conversions were carefully reviewed to ensure compatibility between the Excel dataset and the SQL Server table structure.

Special attention was given to the **OrderDate** column because of date format conversion issues encountered during the initial import attempt. After adjusting the table structure, the dataset was imported successfully.

The import operation transferred all transaction records from the Excel dataset into the Orders table. This step established the foundation for performing SQL-based analysis, data exploration, and business reporting.

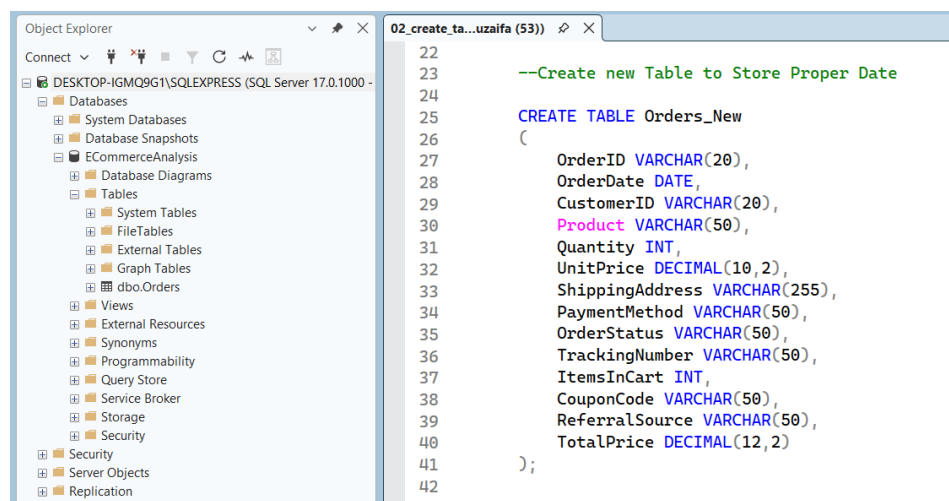


Data Type Conversion and Table Migration

During the data import process, a compatibility issue was encountered with the OrderDate column. The original Orders table

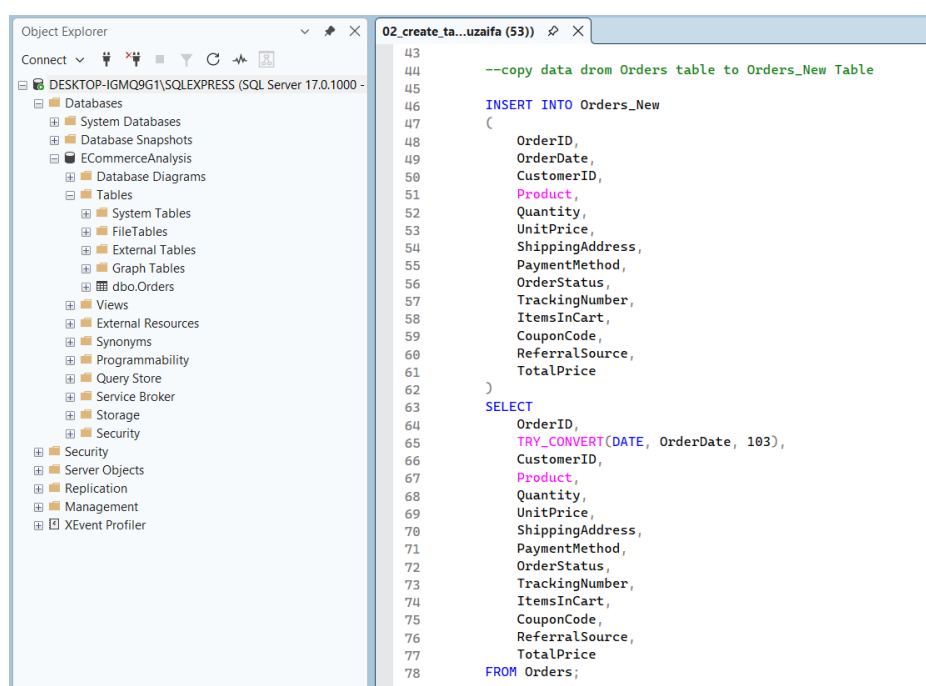
defined the OrderDate field using the DATE data type. However, SQL Server generated conversion errors while importing date values from the Excel dataset due to format inconsistencies.

To resolve this issue, the OrderDate column was temporarily stored as VARCHAR(50) during the import stage. After the dataset was successfully imported, a new table named Orders_New was created with the OrderDate field defined as the DATE data type.



```
22
23
24 --Create new Table to Store Proper Date
25
26 CREATE TABLE Orders_New
27 (
28     OrderID VARCHAR(20),
29     OrderDate DATE,
30     CustomerID VARCHAR(20),
31     Product VARCHAR(50),
32     Quantity INT,
33     UnitPrice DECIMAL(10,2),
34     ShippingAddress VARCHAR(255),
35     PaymentMethod VARCHAR(50),
36     OrderStatus VARCHAR(50),
37     TrackingNumber VARCHAR(50),
38     ItemsInCart INT,
39     CouponCode VARCHAR(50),
40     ReferralSource VARCHAR(50),
41     TotalPrice DECIMAL(12,2)
42 )
43
```

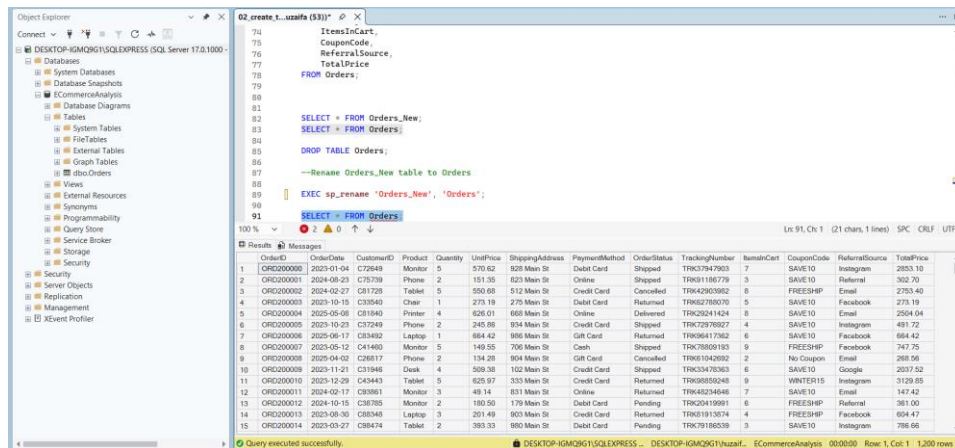
The TRY_CONVERT() function was used to safely convert the imported date values into the DATE format while transferring records from the Orders table to the Orders_New table.



```
43
44
45 --copy data drom Orders table to Orders_New Table
46
47 INSERT INTO Orders_New
48 (
49     OrderID,
50     OrderDate,
51     CustomerID,
52     Product,
53     Quantity,
54     UnitPrice,
55     ShippingAddress,
56     PaymentMethod,
57     OrderStatus,
58     TrackingNumber,
59     ItemsInCart,
60     CouponCode,
61     ReferralSource,
62     TotalPrice
63 )
64 SELECT
65     OrderID,
66     TRY_CONVERT(DATE, OrderDate, 103),
67     CustomerID,
68     Product,
69     Quantity,
70     UnitPrice,
71     ShippingAddress,
72     PaymentMethod,
73     OrderStatus,
74     TrackingNumber,
75     ItemsInCart,
76     CouponCode,
77     ReferralSource,
78     TotalPrice
79 FROM Orders;
80
```

After verifying that all records were transferred successfully, the original Orders table was removed and the Orders_New table was renamed to Orders.

This process ensured that the final database structure contained the correct DATE data type while preserving all imported records.



The screenshot shows a SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' pane displays the database structure, including 'Database Snapshots', 'Databases', 'System Databases', 'Database Diagrams', 'Tables', 'System Tables', 'FileTables', 'External Tables', 'Graph Tables', 'dboOrders', 'Views', 'External Resources', 'Synonyms', 'Programmability', 'Query Store', 'Service Broker', 'Storage', 'Security', 'Server Objects', 'Replication', 'Management', and 'XEvent Profiler'. The main window displays a query window titled '02_create_t_usafra (53)' with the following SQL code:

```
SELECT * FROM Orders_New;
SELECT * FROM Orders;

DROP TABLE Orders;

--Rename Orders_New table to Orders
EXEC sp_rename 'Orders_New', 'Orders';

SELECT * FROM Orders;
```

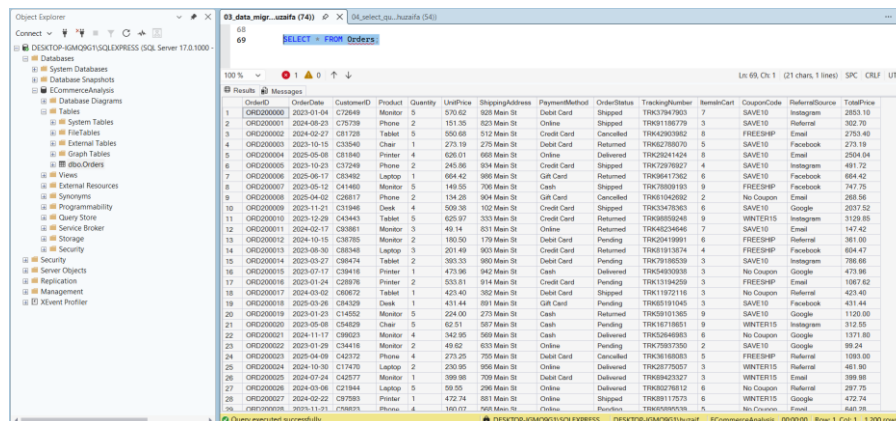
The results pane shows a table with 15 columns: OrderID, OrderDate, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemInCart, CouponCode, ReferralSource, TotalPrice, and TotalPrice. The table contains 15 rows of data, including details for various orders such as ORD200000, ORD200001, ORD200002, etc.

Data Verification

After completing the data import and migration process, a verification query was executed to ensure that all records were successfully transferred to the final Orders table.

The verification process involved displaying sample records from the table and confirming that all columns, including the OrderDate field, contained valid data.

The results confirmed that the dataset was successfully imported and was ready for SQL-based analysis.



The screenshot shows a SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' pane displays the database structure, including 'Database Snapshots', 'Databases', 'System Databases', 'Database Diagrams', 'Tables', 'System Tables', 'FileTables', 'External Tables', 'Graph Tables', 'dboOrders', 'Views', 'External Resources', 'Synonyms', 'Programmability', 'Query Store', 'Service Broker', 'Storage', 'Security', 'Server Objects', 'Replication', 'Management', and 'XEvent Profiler'. The main window displays a query window titled '03_data_migr...usafra (74)' with the following SQL code:

```
SELECT * FROM Orders;
```

The results pane shows a table with 15 columns: OrderID, OrderDate, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemInCart, CouponCode, ReferralSource, TotalPrice, and TotalPrice. The table contains 15 rows of data, including details for various orders such as ORD200000, ORD200001, ORD200002, etc.

SQL Query Analysis

After verifying the imported dataset, several SQL queries were executed to retrieve, filter, sort, group, and summarize data. These queries demonstrate the practical implementation of SQL fundamentals and provide meaningful business insights from the e-commerce dataset.

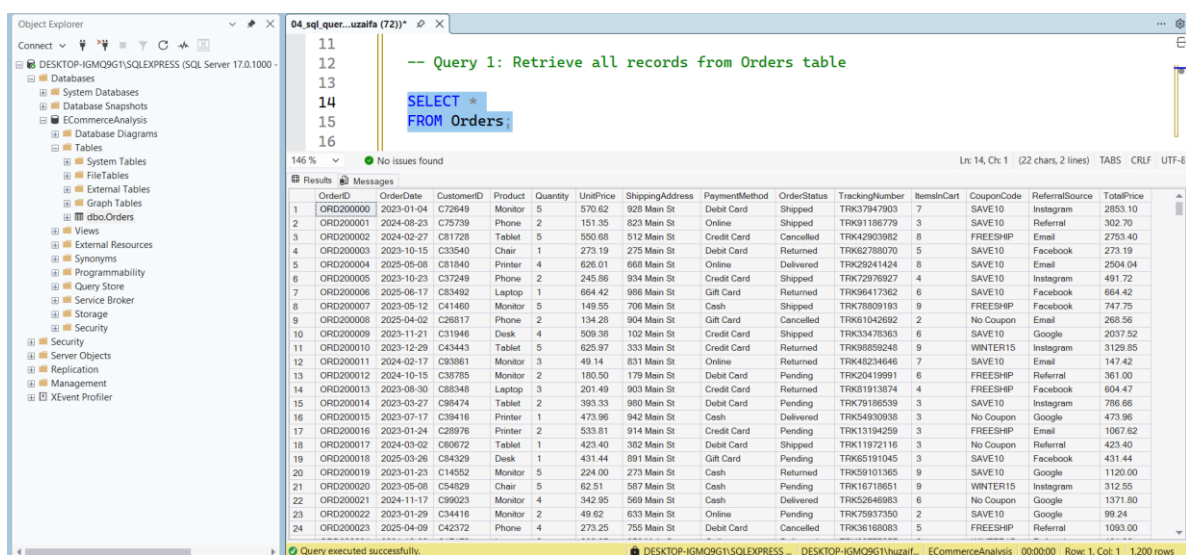
Section 1. SQL Fundamentals (Internship Requirements)

Query 1. Retrieve All Records Using SELECT

The SELECT statement was used to retrieve all records from the Orders table. This query provides a complete view of the dataset and helps analysts understand the available information before performing advanced analysis.

Analysis

The query returned all 1,200 transaction records stored in the Orders table. The output included customer information, product details, order dates, payment methods, order statuses, referral sources, coupon codes, and revenue information.



The screenshot displays the SQL Server Enterprise Manager interface. On the left, the Object Explorer shows the database structure for 'DESKTOP-IGM09G1\SQLEXPRESS (SQL Server 17.0.1000)'. The 'ECommerceAnalysis' database is selected, showing tables like 'Orders'. The central pane shows a query window with the following SQL code:

```
-- Query 1: Retrieve all records from Orders table
SELECT *
FROM Orders;
```

The bottom pane shows the results of the query, displaying 1,200 rows of data. The results are shown in a grid with the following columns: OrderID, OrderDate, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemInCart, CouponCode, ReferralSource, and TotalPrice. The status bar at the bottom indicates 'Query executed successfully.' and '1,200 rows'.

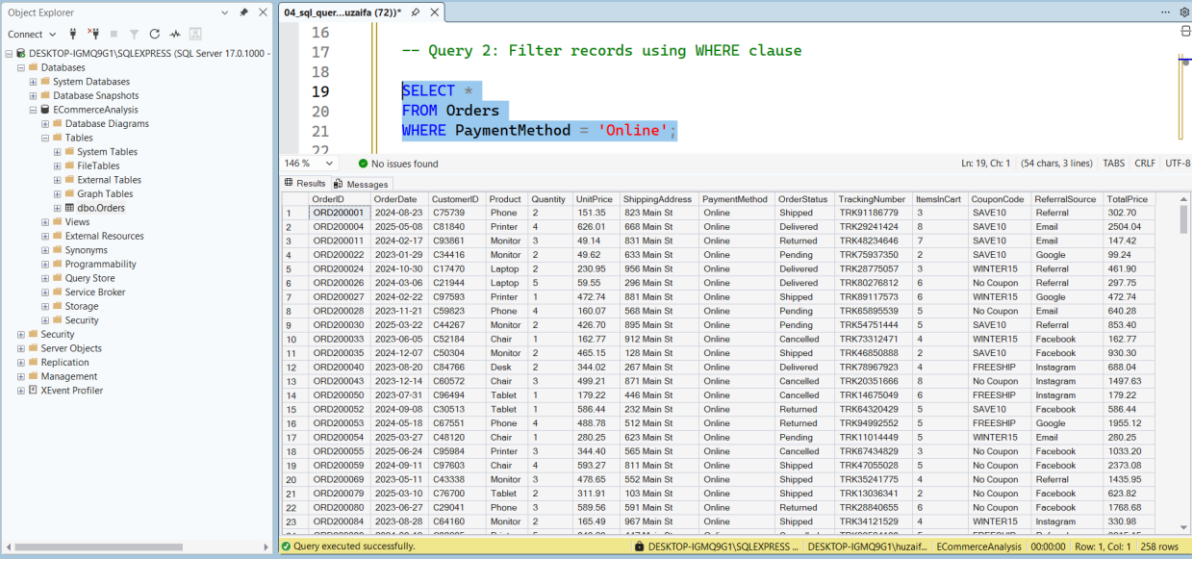
OrderID	OrderDate	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemInCart	CouponCode	ReferralSource	TotalPrice
1	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.10
2	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3		Referral	302.70
3	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.40
4	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62769070	5	SAVE10	Facebook	273.19
5	2025-05-08	C31840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
6	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
7	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
8	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
9	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK81042692	2	No Coupon	Email	268.56
10	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
11	2023-12-29	C43443	Tablet	5	625.97	533 Main St	Credit Card	Returned	TRK98592648	9	WINTER15	Instagram	3129.85
12	2024-02-17	C33861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234648	7	SAVE10	Email	147.42
13	2024-10-15	C38785	Monitor	2	180.50	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361.00
14	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
15	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
16	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	No Coupon	Google	473.96
17	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
18	2024-03-02	C60672	Tablet	1	423.40	382 Main St	Debit Card	Shipped	TRK11972116	3	No Coupon	Referral	423.40
19	2025-03-26	C94329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191945	3	SAVE10	Facebook	431.44
20	2023-01-23	C14552	Monitor	5	224.00	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120.00
21	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
22	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52648983	6	No Coupon	Google	1371.80
23	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
24	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093.00

Query 2. Filter Records Using WHERE Clause

The WHERE clause was used to retrieve only those records that satisfy a specific condition.

Analysis

The query displayed transactions completed using the Online payment method. The results showed a significant number of customers preferred online payments, demonstrating how SQL can be used to filter records based on business requirements.



The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results grid. The query editor displays the following SQL query:

```
-- Query 2: Filter records using WHERE clause  
SELECT *  
FROM Orders  
WHERE PaymentMethod = 'Online'
```

The results grid shows 25 rows of data, sorted by TotalPrice in descending order. The columns are: OrderID, OrderDate, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemshlCart, CouponCode, ReferralSource, and TotalPrice.

OrderID	OrderDate	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemshlCart	CouponCode	ReferralSource	TotalPrice
1	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.70
2	2025-05-08	C81940	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
3	2024-02-17	C33661	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
4	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
5	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.90
6	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	No Coupon	Referral	297.75
7	2024-02-22	C87593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74
8	2023-11-21	C59523	Phone	4	160.07	568 Main St	Online	Pending	TRK65895539	5	No Coupon	Email	640.28
9	2025-03-22	C44267	Monitor	2	426.70	895 Main St	Online	Pending	TRK54751444	5	SAVE10	Referral	853.40
10	2025-06-05	C32184	Chair	1	162.77	912 Main St	Online	Cancelled	TRK73312471	4	WINTER15	Facebook	162.77
11	2024-12-07	C50304	Monitor	2	465.15	126 Main St	Online	Shipped	TRK46850888	2	SAVE10	Facebook	930.30
12	2023-08-20	C84766	Desk	2	344.02	267 Main St	Online	Delivered	TRK78967923	4	FREESHIP	Instagram	688.04
13	2023-12-14	C60572	Chair	3	499.21	871 Main St	Online	Cancelled	TRK20351666	8	No Coupon	Instagram	1497.63
14	2023-07-31	C96494	Tablet	1	179.22	446 Main St	Online	Cancelled	TRK14675049	6	FREESHIP	Instagram	179.22
15	2024-09-08	C30513	Tablet	1	586.44	232 Main St	Online	Returned	TRK64320429	5	SAVE10	Facebook	586.44
16	2024-05-15	C67551	Phone	4	488.78	512 Main St	Online	Returned	TRK94992552	5	FREESHIP	Google	1955.12
17	2025-03-27	C48120	Chair	1	280.25	623 Main St	Online	Pending	TRK11014449	5	WINTER15	Email	280.25
18	2025-06-24	C95864	Printer	3	344.40	565 Main St	Online	Cancelled	TRK67434629	3	No Coupon	Facebook	1033.20
19	2024-09-11	C87603	Chair	4	593.27	811 Main St	Online	Shipped	TRK47055028	5	No Coupon	Facebook	2373.08
20	2023-05-11	C43338	Monitor	3	478.65	552 Main St	Online	Shipped	TRK35241775	4	No Coupon	Referral	1435.95
21	2025-03-10	C76700	Tablet	2	311.91	103 Main St	Online	Shipped	TRK13036341	2	No Coupon	Facebook	623.82
22	2023-06-27	C29041	Phone	3	589.56	591 Main St	Online	Returned	TRK28840655	6	No Coupon	Facebook	1768.68
23	2023-08-28	C64160	Monitor	2	165.49	967 Main St	Online	Shipped	TRK34121529	4	WINTER15	Instagram	330.98

Query 3. Sort Records Using ORDER BY Clause

The ORDER BY clause was used to sort records based on TotalPrice in descending order.

Analysis

The query arranged transactions from the highest-value order to the lowest-value order. The highest transaction value observed in the dataset was approximately **3456.40**, helping identify premium customer purchases and high-revenue transactions.

04_sql_query...uzaifa (72) ✕

```
-- Query 3: Sort records using ORDER BY clause

SELECT *
FROM Orders
ORDER BY TotalPrice DESC;
```

146 % No issues found Lrn: 25, Cht: 1 (48 chars, 3 lines) TABS CRLF UTF-8

	OrderID	OrderDate	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
1	ORD200788	2023-08-17	C57276	Tablet	5	691.28	183 Main St	Online	Delivered	TRK75899752	10	SAVE10	Email	3456.40
2	ORD201122	2023-06-07	C38840	Monitor	5	678.19	766 Main St	Online	Returned	TRK32496970	8	No Coupon	Facebook	3390.95
3	ORD200632	2023-05-02	C67260	Laptop	5	678.16	463 Main St	Gift Card	Delivered	TRK38229104	7	WINTER15	Facebook	3390.80
4	ORD200469	2023-11-26	C13877	Chair	5	676.98	893 Main St	Cash	Cancelled	TRK17254691	5	No Coupon	Facebook	3384.90
5	ORD200328	2023-02-28	C18404	Tablet	5	674.04	546 Main St	Online	Cancelled	TRK89401624	7	SAVE10	Google	3370.20
6	ORD200107	2023-03-27	C16775	Printer	5	670.75	848 Main St	Gift Card	Shipped	TRK34392124	8	FREESHIP	Instagram	3353.75
7	ORD200326	2024-07-01	C65986	Laptop	5	670.48	273 Main St	Gift Card	Returned	TRK98353867	5	SAVE10	Facebook	3352.40
8	ORD201065	2023-10-30	C47778	Printer	5	666.80	488 Main St	Debit Card	Delivered	TRK79504329	7	SAVE10	Referral	3334.00
9	ORD201031	2023-02-28	C59183	Phone	5	664.51	136 Main St	Debit Card	Pending	TRK18129706	8	SAVE10	Email	3322.55
10	ORD200463	2023-05-26	C25276	Laptop	5	662.78	214 Main St	Debit Card	Shipped	TRK14011732	9	SAVE10	Instagram	3313.90
11	ORD200361	2024-06-29	C53464	Printer	5	659.85	197 Main St	Credit Card	Delivered	TRK32743531	9	SAVE10	Facebook	3299.25
12	ORD200367	2024-04-25	C13108	Laptop	5	658.77	712 Main St	Gift Card	Pending	TRK82284400	10	FREESHIP	Instagram	3293.85
13	ORD200837	2025-06-14	C88029	Chair	5	655.55	282 Main St	Online	Delivered	TRK59701278	5	No Coupon	Email	3277.75
14	ORD200527	2024-08-19	C27202	Chair	5	653.47	967 Main St	Credit Card	Cancelled	TRK75406128	10	SAVE10	Referral	3267.35
15	ORD200768	2023-09-04	C35987	Tablet	5	653.46	165 Main St	Cash	Cancelled	TRK28918815	9	WINTER15	Email	3267.30
16	ORD200889	2024-10-27	C36725	Monitor	5	650.72	350 Main St	Credit Card	Cancelled	TRK16808790	8	WINTER15	Email	3253.60
17	ORD200540	2024-01-29	C87281	Laptop	5	648.65	852 Main St	Online	Pending	TRK81765057	7	No Coupon	Referral	3243.25
18	ORD200802	2024-02-17	C84331	Chair	5	644.64	205 Main St	Gift Card	Cancelled	TRK34440714	6	FREESHIP	Email	3223.20
19	ORD200957	2023-01-12	C73575	Monitor	5	643.89	851 Main St	Cash	Returned	TRK23387612	8	WINTER15	Facebook	3219.45
20	ORD200086	2024-06-19	C88205	Printer	5	643.03	447 Main St	Online	Cancelled	TRK93504166	5	FREESHIP	Referral	3215.15
21	ORD200221	2024-02-23	C83473	Tablet	5	639.37	552 Main St	Debit Card	Pending	TRK50003889	6	SAVE10	Google	3196.85
22	ORD200967	2025-05-06	C52879	Tablet	5	638.92	502 Main St	Credit Card	Returned	TRK67403284	5	FREESHIP	Google	3194.60
23	ORD200296	2024-06-19	C48453	Desk	5	638.80	434 Main St	Debit Card	Cancelled	TRK52110494	10	WINTER15	Referral	3194.00

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzairf... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 1,200 rows

Query 4. Product Analysis Using GROUP BY

The GROUP BY clause was used to group records by product and calculate total orders for each product.

Analysis

The query summarized product demand across all transactions.

Based on the analysis, **Printer** was the most frequently ordered product with **181 orders**, while **Phone** received the lowest number of orders with **156 orders**.

04_sql_query...uzaifa (72)* ✕

```
-- Query 4: Product analysis using GROUP BY

SELECT Product,
COUNT(*) AS TotalOrders
FROM Orders
GROUP BY Product
ORDER BY TotalOrders DESC;
```

146 % No issues found Lrn: 31, Cht: 1 (99 chars, 5 lines) TABS CRLF UTF-8

	Product	TotalOrders
1	Printer	181
2	Tablet	179
3	Chair	178
4	Laptop	173
5	Desk	170
6	Monitor	163
7	Phone	156

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzairf... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 7 rows

Aggregate Function Analysis

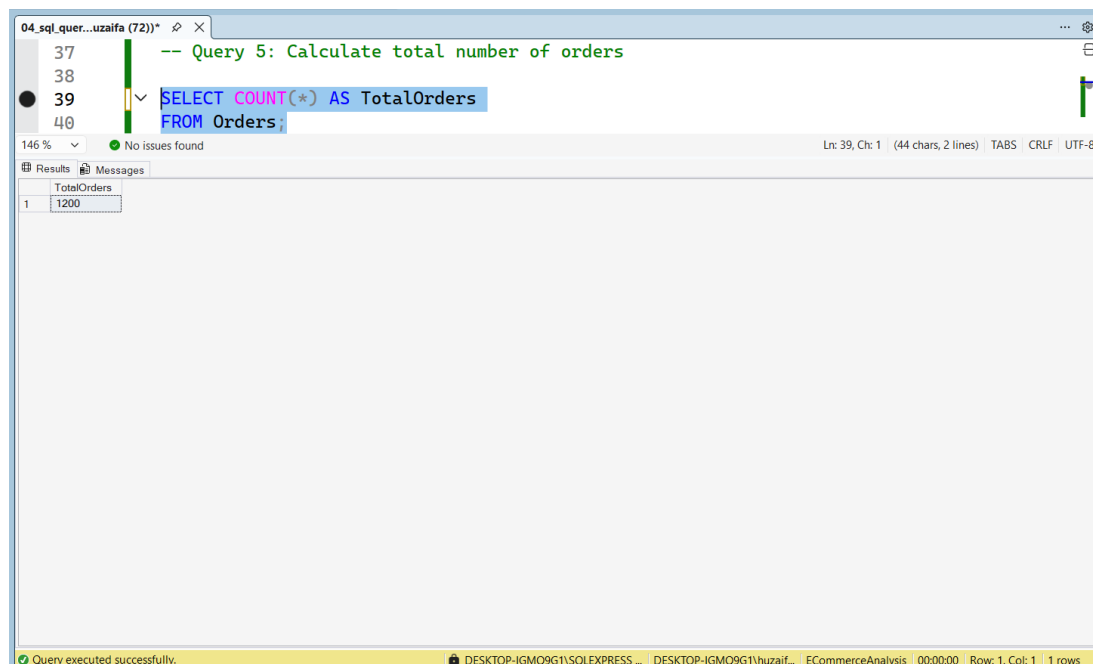
Aggregate functions were used to summarize large amounts of data and generate important business metrics.

Query 5. Total Orders Using COUNT()

The COUNT() function was used to calculate the total number of orders present in the dataset.

Analysis

The query returned a total of **1,200 orders**, confirming that all records were successfully imported and available for analysis.



Query 6. Total Revenue Using SUM()

The SUM() function was used to calculate total revenue generated from all customer transactions.

Analysis

The query returned a total revenue of **1,264,761.96**. This value represents the overall sales generated from all e-commerce transactions included in the dataset.

```
04_sql_quer...uzaiifa (72)) *  
42 -- Query 6: Calculate total revenue  
43  
44 SELECT SUM(TotalPrice) AS TotalRevenue  
45 FROM Orders;
```

146 % No issues found Ln: 44, Ch: 1 (52 chars, 2 lines) TABS CRLF UTF-8

	TotalRevenue
1	1264761.96

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzaiif... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 1 rows

Query 7. Average Order Value Using AVG()

The AVG() function was used to calculate the average amount spent per order.

Analysis

The query returned an average order value of **1053.97**. This indicates that customers spent approximately **1,054 currency units per transaction** on average.

```
04_sql_quer...uzaiifa (72)) *  
47 -- Query 7: Calculate average order value  
48  
49 SELECT AVG(TotalPrice) AS AverageOrderValue  
50 FROM Orders;
```

146 % No issues found Ln: 49, Ch: 1 (57 chars, 2 lines) TABS CRLF UTF-8

	AverageOrderValue
1	1053.968300

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzaiif... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 1 rows

Section 2. Intermediate SQL Practice

After completing the fundamental SQL operations, additional queries were executed to practice intermediate-level SQL concepts. These queries demonstrate the use of logical operators, filtering techniques, pattern matching, distinct values, and additional aggregate functions.

Query 8. Filter Records Using AND Condition

The AND operator was used to retrieve records where multiple conditions were satisfied simultaneously.

Analysis

The query displayed transactions where customers used the **Online** payment method and the order status was **Delivered**. The result returned **63 records**, indicating successfully completed online transactions. This analysis helps evaluate the effectiveness of online payment processing and order fulfillment.

04_sql_query...uzaifa (68) ✕

52

-- =====

53

-- SECTION 2 - INTERMEDIATE SQL PRACTICE

54

-- Additional Queries

55

-- =====

56

57

-- Query 8: Filter records using AND condition

58

59

SELECT *

60

FROM Orders

61

WHERE PaymentMethod = 'Online'

62

AND OrderStatus = 'Delivered';

146 %

No issues found

Ln: 59, Ch: 1 (85 chars, 4 lines)

TABS

CRLF

UTF-8

Results Messages

	OrderID	OrderDate	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
1	ORD2000004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
2	ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.90
3	ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	No Coupon	Referral	297.75
4	ORD200040	2023-08-20	C84766	Desk	2	344.02	267 Main St	Online	Delivered	TRK78967923	4	FREESHIP	Instagram	688.04
5	ORD200100	2023-02-17	C74866	Phone	3	664.79	743 Main St	Online	Delivered	TRK53783265	6	No Coupon	Facebook	1994.37
6	ORD200123	2023-07-06	C86064	Desk	3	278.04	727 Main St	Online	Delivered	TRK62182901	5	No Coupon	Instagram	834.12
7	ORD200140	2025-05-31	C33110	Chair	3	528.14	388 Main St	Online	Delivered	TRK38624225	3	FREESHIP	Email	1584.42
8	ORD200156	2024-11-08	C58808	Chair	2	269.83	651 Main St	Online	Delivered	TRK24451893	4	SAVE10	Email	539.66
9	ORD200169	2024-04-07	C35785	Printer	1	427.82	732 Main St	Online	Delivered	TRK78923089	2	WINTER15	Email	427.82
10	ORD200172	2025-04-06	C34134	Printer	1	254.12	365 Main St	Online	Delivered	TRK43074303	1	WINTER15	Email	254.12
11	ORD200173	2024-03-24	C95901	Phone	3	664.46	806 Main St	Online	Delivered	TRK56989066	4	FREESHIP	Google	1993.38
12	ORD200178	2024-04-11	C53070	Chair	4	512.06	283 Main St	Online	Delivered	TRK49791755	5	SAVE10	Google	2048.24
13	ORD200214	2024-08-21	C29483	Desk	4	305.72	673 Main St	Online	Delivered	TRK89423803	5	No Coupon	Facebook	1222.88
14	ORD200254	2023-12-26	C10866	Monitor	3	263.19	770 Main St	Online	Delivered	TRK34809583	5	No Coupon	Instagram	789.57
15	ORD200285	2025-06-10	C67583	Phone	1	338.31	974 Main St	Online	Delivered	TRK93281700	3	No Coupon	Google	338.31
16	ORD200307	2023-07-01	C92407	Laptop	3	557.41	135 Main St	Online	Delivered	TRK11755610	4	No Coupon	Email	1672.23
17	ORD200314	2023-10-04	C29931	Desk	4	153.09	540 Main St	Online	Delivered	TRK48712417	4	FREESHIP	Referral	612.36
18	ORD200331	2023-04-15	C78724	Printer	2	422.13	187 Main St	Online	Delivered	TRK56203134	7	SAVE10	Facebook	1366.30

Query executed successfully.

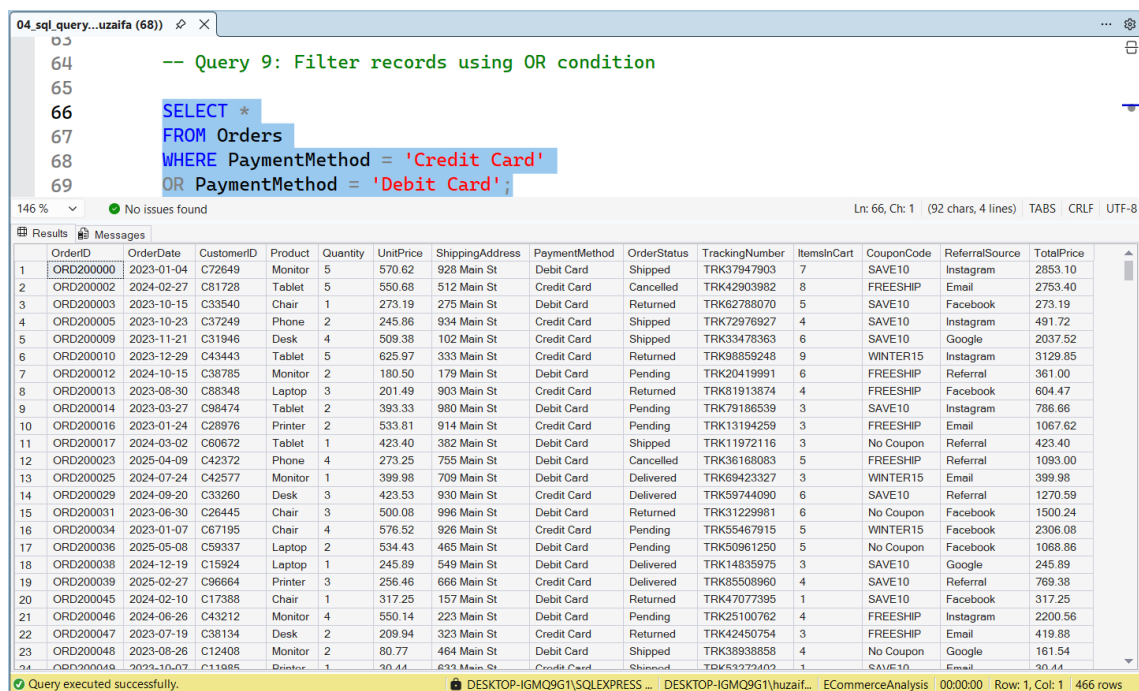
DESKTOP-IGM09G1\SQLEXPRESS ... DESKTOP-IGM09G1\huzai... ECommerceAnalysis 00:00:00 Row: 1. Col: 1 63 rows

Query 9. Filter Records Using OR Condition

The OR operator was used to retrieve records matching either of the specified conditions.

Analysis

The query displayed orders processed through either **Credit Card** or **Debit Card** payment methods. The result returned **466 records**, showing that card-based payments represent a significant portion of customer transactions.



```
-- Query 9: Filter records using OR condition

SELECT *
FROM Orders
WHERE PaymentMethod = 'Credit Card'
OR PaymentMethod = 'Debit Card';
```

	OrderID	OrderDate	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
1	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.10
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.40
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
4	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
5	ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
6	ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
7	ORD200012	2024-10-15	C38785	Monitor	2	180.50	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361.00
8	ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
9	ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
10	ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
11	ORD200017	2024-03-02	C80872	Tablet	1	423.40	382 Main St	Debit Card	Shipped	TRK11972116	3	No Coupon	Referral	423.40
12	ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093.00
13	ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
14	ORD200029	2024-09-20	C33260	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRK59744090	6	SAVE10	Referral	1270.59
15	ORD200031	2023-06-30	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6	No Coupon	Facebook	1500.24
16	ORD200034	2023-01-07	C67195	Chair	4	576.52	926 Main St	Credit Card	Pending	TRK55467915	5	WINTER15	Facebook	2306.08
17	ORD200036	2025-05-08	C59337	Laptop	2	534.43	465 Main St	Debit Card	Pending	TRK50961250	5	No Coupon	Facebook	1068.86
18	ORD200038	2024-12-19	C15924	Laptop	1	245.89	549 Main St	Debit Card	Delivered	TRK14835975	3	SAVE10	Google	245.89
19	ORD200039	2025-02-27	C96664	Printer	3	256.46	666 Main St	Credit Card	Delivered	TRK85508960	4	SAVE10	Referral	769.38
20	ORD200045	2024-02-10	C17388	Chair	1	317.25	157 Main St	Debit Card	Returned	TRK47077395	1	SAVE10	Facebook	317.25
21	ORD200046	2024-06-26	C43212	Monitor	4	550.14	223 Main St	Debit Card	Pending	TRK25100762	4	FREESHIP	Instagram	2200.56
22	ORD200047	2023-07-19	C38134	Desk	2	209.94	323 Main St	Credit Card	Returned	TRK42450754	3	FREESHIP	Email	419.88
23	ORD200048	2023-08-26	C12408	Monitor	2	80.77	464 Main St	Debit Card	Shipped	TRK38938858	4	No Coupon	Google	161.54
24	ORD200049	2023-10-07	C41085	Printer	1	30.44	633 Main St	Credit Card	Shipped	TRK53272402	1	SAVE10	Email	30.44

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzai... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 466 rows

Query 10. Filter Records Using BETWEEN

The BETWEEN operator was used to retrieve orders whose TotalPrice falls within a specific range.

Analysis

The query returned **495 records** with order values between **500 and 1500**. This analysis helps identify medium-value purchases and provides insight into common customer spending patterns.

04_sql_query...uzaifa (68)

```

71 -- Query 10: Filter records using BETWEEN
72
73 SELECT *
74 FROM Orders
75 WHERE TotalPrice BETWEEN 500 AND 1500;
76

```

146 % No issues found Ln: 73, Ch: 1 (61 chars, 3 lines) TABS CRLF UTF-8

	OrderID	OrderDate	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
1	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
2	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
3	ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
4	ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
5	ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
6	ORD200019	2024-11-23	C14552	Monitor	5	224.00	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120.00
7	ORD200021	2024-11-17	C98023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	No Coupon	Google	1371.80
8	ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093.00
9	ORD200028	2023-11-21	C59823	Phone	4	160.07	568 Main St	Online	Pending	TRK65895539	5	No Coupon	Email	640.28
10	ORD200029	2024-09-20	C33260	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRK59744090	6	SAVE10	Referral	1270.59
11	ORD200030	2025-03-22	C44267	Monitor	2	426.70	895 Main St	Online	Pending	TRK54751444	5	SAVE10	Referral	853.40
12	ORD200035	2024-12-07	C50304	Monitor	2	465.15	128 Main St	Online	Shipped	TRK46850888	2	SAVE10	Facebook	930.30
13	ORD200036	2025-05-08	C59337	Laptop	2	534.43	465 Main St	Debit Card	Pending	TRK50961250	5	No Coupon	Facebook	1068.86
14	ORD200039	2025-02-27	C96664	Printer	3	256.46	666 Main St	Credit Card	Delivered	TRK85508960	4	SAVE10	Referral	769.38
15	ORD200040	2023-08-20	C94766	Desk	2	344.02	267 Main St	Online	Delivered	TRK78967923	4	FREESHIP	Instagram	688.04
16	ORD200043	2023-12-14	C60572	Chair	3	499.21	871 Main St	Online	Cancelled	TRK20351666	8	No Coupon	Instagram	1497.63
17	ORD200044	2025-04-21	C73225	Phone	2	550.18	996 Main St	Gift Card	Returned	TRK73848522	3	FREESHIP	Referral	1100.36
18	ORD200052	2024-09-08	C30513	Tablet	1	586.44	232 Main St	Online	Returned	TRK64320429	5	SAVE10	Facebook	586.44
19	ORD200055	2025-06-24	C95984	Printer	3	344.40	565 Main St	Online	Cancelled	TRK87434829	3	No Coupon	Facebook	1033.20
20	ORD200056	2023-05-21	C54298	Tablet	3	479.21	566 Main St	Cash	Cancelled	TRK62802341	5	WINTER15	Referral	1437.63
21	ORD200063	2025-02-02	C96895	Chair	1	564.86	813 Main St	Debit Card	Shipped	TRK64283851	6	WINTER15	Referral	564.86
22	ORD200066	2024-11-15	C18104	Chair	5	127.18	715 Main St	Cash	Cancelled	TRK15488568	9	SAVE10	Referral	635.90
23	ORD200068	2023-12-11	C41596	Chair	3	480.92	709 Main St	Credit Card	Pending	TRK31665940	5	FREESHIP	Email	1472.76
24	ORD200069	2025-05-11	C19338	Monitor	3	478.65	569 Main St	Online	Shipped	TRK789411736	4	No Coupon	Referral	1435.95

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzair... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 495 rows

Query 11. Filter Records Using IN Operator

The IN operator was used to retrieve records belonging to selected products.

Analysis

The query displayed orders related to **Laptop**, **Phone**, and **Monitor** products. A total of **492 records** were returned, demonstrating how the IN operator simplifies filtering multiple values within a single condition.

04_sql_query...uzaifa (68)

```

77 -- Query 11: Filter records using IN operator
78
79 SELECT *
80 FROM Orders
81 WHERE Product IN ('Laptop', 'Phone', 'Monitor');
82

```

146 % No issues found Ln: 79, Ch: 1 (69 chars, 3 lines) TABS CRLF UTF-8

	OrderID	OrderDate	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
1	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.10
2	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.70
3	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
4	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
5	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
6	ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	No Coupon	Email	268.56
7	ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
8	ORD200012	2024-10-15	C38785	Monitor	2	180.50	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361.00
9	ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
10	ORD200019	2023-01-23	C14552	Monitor	5	224.00	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120.00
11	ORD200021	2024-11-17	C98023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	No Coupon	Google	1371.80
12	ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
13	ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093.00
14	ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.90
15	ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
16	ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	No Coupon	Referral	297.75
17	ORD200028	2023-11-21	C59823	Phone	4	160.07	568 Main St	Online	Pending	TRK65895539	5	No Coupon	Email	640.28
18	ORD200030	2025-03-22	C44267	Monitor	2	426.70	895 Main St	Online	Pending	TRK54751444	5	SAVE10	Referral	853.40
19	ORD200035	2024-12-07	C50304	Monitor	2	465.15	128 Main St	Online	Shipped	TRK46850888	2	SAVE10	Facebook	930.30
20	ORD200036	2025-05-08	C59337	Laptop	2	534.43	465 Main St	Debit Card	Pending	TRK50961250	5	No Coupon	Facebook	1068.86
21	ORD200037	2024-10-14	C10054	Laptop	2	76.74	409 Main St	Cash	Returned	TRK94005261	7	SAVE10	Instagram	153.48
22	ORD200038	2024-12-19	C15924	Laptop	1	245.89	549 Main St	Debit Card	Delivered	TRK14835975	3	SAVE10	Google	245.89
23	ORD200041	2023-12-05	C20212	Monitor	3	611.45	657 Main St	Gift Card	Pending	TRK69733327	4	FREESHIP	Instagram	1834.35
24	ORD200044	2025-04-21	C73225	Phone	2	550.18	996 Main St	Gift Card	Returned	TRK73848522	3	FREESHIP	Referral	1100.36

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzair... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 492 rows

Query 12. Search Records Using LIKE Operator

The LIKE operator was used for pattern-based searching.

Analysis

The query retrieved customers whose CustomerID starts with "C1". The result returned **146 records**, demonstrating the use of pattern matching for text-based searches and customer segmentation.

```
04_sql_query...uzaiifa (68) x
83 -- Query 12: Search records using LIKE operator
84
85 SELECT *
86 FROM Orders
87 WHERE CustomerID LIKE 'C1%';
88
```

146 % No issues found Ln: 85, Ch: 1 (51 chars, 3 lines) TABS CRLF UTF-8

	OrderID	OrderDate	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
1	ORD200019	2023-01-23	C14552	Monitor	5	224.00	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120.00
2	ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.90
3	ORD200032	2023-05-01	C12388	Tablet	5	536.72	830 Main St	Cash	Delivered	TRK99261395	10	No Coupon	Email	2683.60
4	ORD200037	2024-10-14	C10054	Laptop	2	76.74	409 Main St	Cash	Returned	TRK94005261	7	SAVE10	Instagram	153.48
5	ORD200038	2024-12-19	C15924	Laptop	1	245.89	549 Main St	Debit Card	Delivered	TRK14835975	3	SAVE10	Google	245.89
6	ORD200045	2024-02-10	C17388	Chair	1	317.25	157 Main St	Debit Card	Returned	TRK47077395	1	SAVE10	Facebook	317.25
7	ORD200048	2023-08-26	C12408	Monitor	2	80.77	464 Main St	Debit Card	Shipped	TRK38938858	4	No Coupon	Google	161.54
8	ORD200049	2023-10-07	C11985	Printer	1	30.44	633 Main St	Credit Card	Shipped	TRK53272402	1	SAVE10	Email	30.44
9	ORD200066	2024-11-15	C18104	Chair	5	127.18	715 Main St	Cash	Cancelled	TRK15488568	9	SAVE10	Referral	635.90
10	ORD200072	2024-12-22	C19451	Tablet	4	589.44	942 Main St	Gift Card	Returned	TRK87787793	5	No Coupon	Facebook	2357.76
11	ORD200073	2023-05-03	C17269	Monitor	4	433.45	129 Main St	Gift Card	Cancelled	TRK79608389	4	No Coupon	Facebook	1733.80
12	ORD200083	2023-08-22	C15681	Laptop	4	540.41	989 Main St	Credit Card	Cancelled	TRK75112172	6	No Coupon	Referral	2161.64
13	ORD200085	2024-09-09	C14553	Tablet	3	146.73	768 Main St	Cash	Delivered	TRK93538334	8	FREESHIP	Email	440.19
14	ORD200089	2025-02-25	C17181	Printer	5	455.33	291 Main St	Credit Card	Cancelled	TRK78912978	8	FREESHIP	Google	2276.65
15	ORD200097	2025-01-21	C15624	Tablet	5	619.52	922 Main St	Debit Card	Pending	TRK40621141	9	No Coupon	Email	3097.60
16	ORD200101	2023-11-25	C15283	Printer	1	381.02	152 Main St	Cash	Shipped	TRK19286291	6	No Coupon	Google	381.02
17	ORD200102	2023-03-08	C18037	Laptop	2	70.34	944 Main St	Gift Card	Cancelled	TRK47362236	3	FREESHIP	Email	140.68
18	ORD200107	2023-03-27	C16775	Printer	5	670.75	848 Main St	Gift Card	Shipped	TRK34392124	8	FREESHIP	Instagram	3353.75
19	ORD200119	2025-06-11	C14683	Desk	2	534.67	187 Main St	Debit Card	Shipped	TRK47373682	7	FREESHIP	Referral	1069.34
20	ORD200134	2025-06-23	C16253	Laptop	2	199.70	583 Main St	Cash	Pending	TRK17354894	4	No Coupon	Email	399.40
21	ORD200182	2025-03-19	C16612	Desk	1	600.54	596 Main St	Debit Card	Pending	TRK96188223	4	WINTER15	Google	600.54
22	ORD200193	2023-07-21	C11644	Printer	1	46.66	429 Main St	Cash	Cancelled	TRK81320339	6	No Coupon	Google	46.66
23	ORD200201	2023-06-18	C10126	Monitor	2	674.90	146 Main St	Online	Shipped	TRK75607531	4	SAVE10	Facebook	1349.80
24	ORD200205	2023-03-04	C19266	Desk	3	147.39	600 Main St	Debit Card	Returned	TRK58357269	3	No Coupon	Email	442.17

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzai... ECommerceAnalysis 00:00:00 Row: 16, Col: 1 146 rows

Query 13. Display Unique Payment Methods

The DISTINCT keyword was used to display unique payment methods.

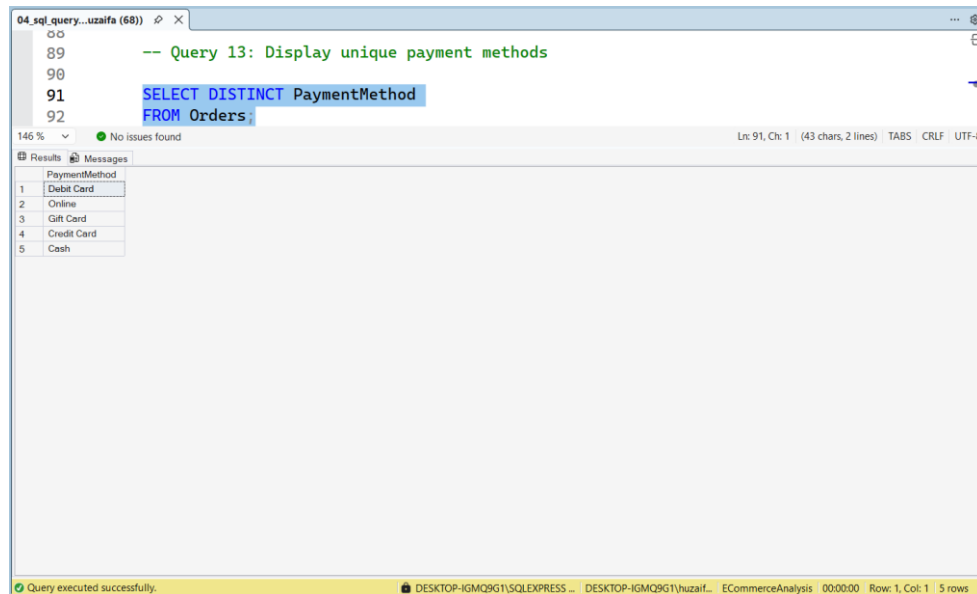
Analysis

The query returned **5 unique payment methods** available in the dataset:

- Debit Card
- Online
- Gift Card
- Credit Card

- Cash

This analysis helps understand the variety of payment options offered to customers.



```
04_sql_query...uzaifa (68)
89  -- Query 13: Display unique payment methods
90
91  SELECT DISTINCT PaymentMethod
92  FROM Orders;
```

PaymentMethod
1 Debit Card
2 Online
3 Gift Card
4 Credit Card
5 Cash

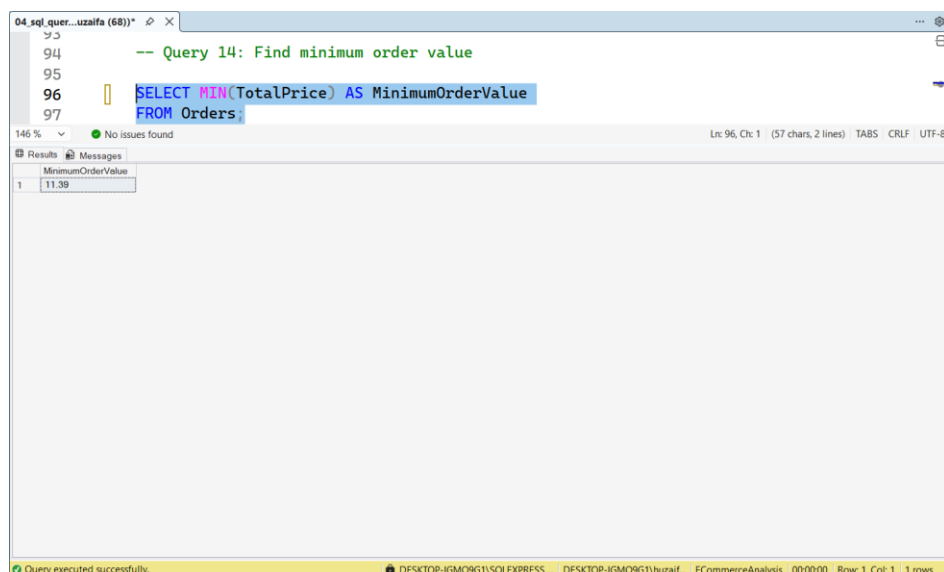
Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzai... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 5 rows

Query 14. Find Minimum Order Value

The MIN() function was used to identify the smallest transaction value.

Analysis

The query returned a minimum order value of **11.39**, representing the lowest customer transaction recorded in the dataset.



```
04_sql_quer...uzaifa (68)*
94  -- Query 14: Find minimum order value
95
96  SELECT MIN(TotalPrice) AS MinimumOrderValue
97  FROM Orders;
```

MinimumOrderValue
1 11.39

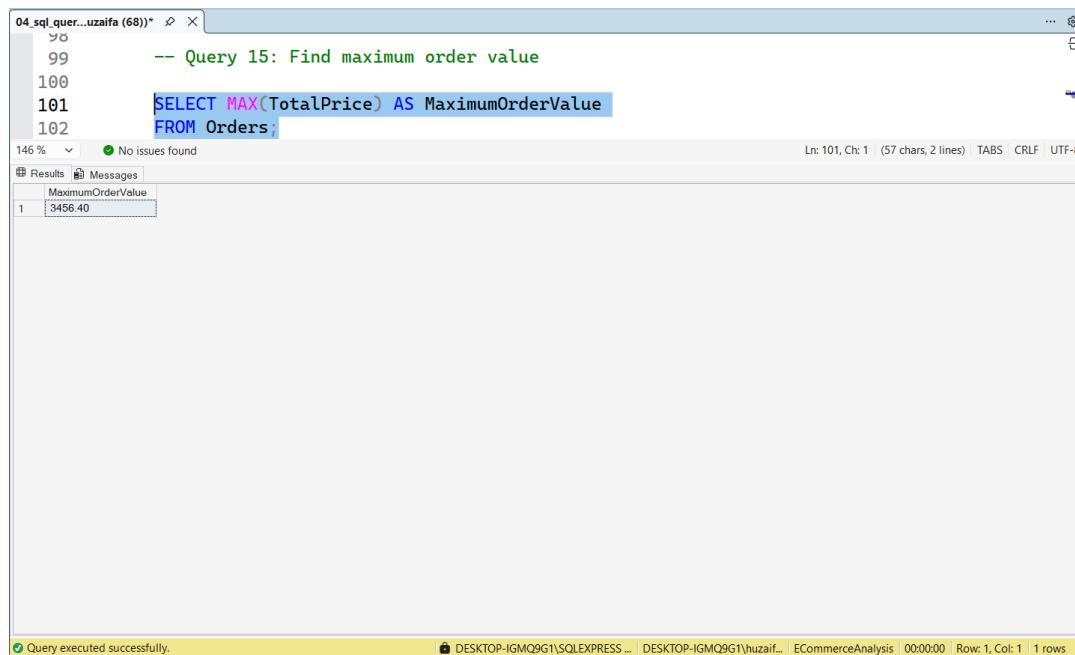
Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzai... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 1 rows

Query 15. Find Maximum Order Value

The MAX() function was used to identify the highest transaction value.

Analysis

The query returned a maximum order value of **3456.40**, representing the highest-value transaction in the dataset. This helps identify premium purchases and high-spending customers.



```
04_sql_quer...uzaifa (68))*
99      -- Query 15: Find maximum order value
100
101      SELECT MAX(TotalPrice) AS MaximumOrderValue
102      FROM Orders;
```

146 % No issues found Ln: 101, Ch: 1 (57 chars, 2 lines) TABS CRLF UTF-8

MaximumOrderValue
3456.40

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzai... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 1 rows

Section 3. Business Analysis

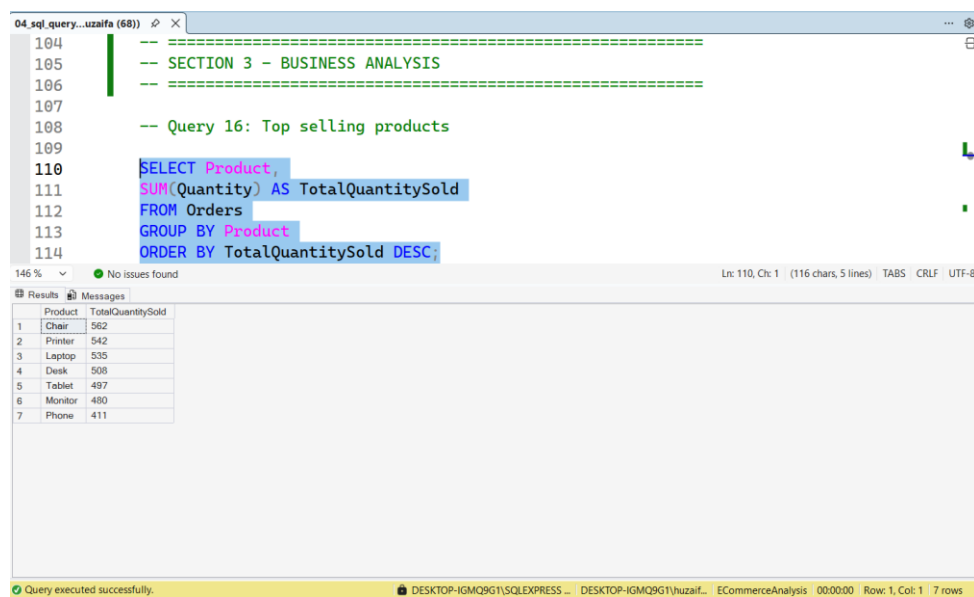
After learning SQL fundamentals and intermediate concepts, business-focused queries were executed to generate meaningful insights from the e-commerce dataset. These queries helped evaluate product performance, revenue generation, payment trends, referral effectiveness, and customer purchasing behavior.

Query 16. Top Selling Products

The query was used to calculate the total quantity sold for each product.

Analysis

The results showed that **Chair** was the top-selling product with a total quantity sold of **562 units**, followed by **Printer (542 units)** and **Laptop (535 units)**. On the other hand, **Phone** recorded the lowest sales volume with **411 units**. This analysis helps identify customer preferences and supports inventory planning and product demand forecasting.



The screenshot shows a SQL query editor window titled "04.sql.query...uzaifa (68)". The query is as follows:

```
-- =====  
-- SECTION 3 - BUSINESS ANALYSIS  
-- =====  
  
-- Query 16: Top selling products  
  
SELECT Product,  
SUM(Quantity) AS TotalQuantitySold  
FROM Orders  
GROUP BY Product  
ORDER BY TotalQuantitySold DESC;
```

The results are displayed in a table with the following data:

Product	TotalQuantitySold
Chair	562
Printer	542
Laptop	535
Desk	508
Tablet	497
Monitor	480
Phone	411

The status bar at the bottom indicates "Query executed successfully." and "DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\juzail... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 | 7 rows".

Query 17. Product Revenue Analysis

The query was used to calculate revenue generated by each product.

Analysis

The results indicated that **Chair** generated the highest revenue of **195,620.11**, closely followed by **Printer (195,612.61)** and **Laptop (192,126.56)**. **Phone** generated the lowest revenue at **151,722.39**. This analysis helps identify high-performing products and supports pricing, inventory, and promotional decisions.

```
115
116      -- Query 17: Product revenue analysis
117
118      SELECT Product,
119             SUM(TotalPrice) AS ProductRevenue
120      FROM Orders
121      GROUP BY Product
122      ORDER BY ProductRevenue DESC;
```

146 % No issues found Lr: 118, Ch: 1 (112 chars, 5 lines) TABS CRLF UTF-8

Product	ProductRevenue
Chair	195620.11
Printer	195612.61
Laptop	192126.56
Tablet	186568.95
Monitor	175651.41
Desk	167459.93
Phone	151722.39

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzalf... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 7 rows

Query 18. Payment Method Analysis

The query was used to calculate the number of transactions completed using each payment method.

Analysis

The results showed that **Online Payment** was the most frequently used payment method with **258 transactions**, followed by **Cash (246 transactions)**. **Gift Card** recorded the lowest number of transactions with **230 orders**. This information helps businesses understand customer payment preferences and optimize payment options.

```
123
124      -- Query 18: Payment method analysis
125
126      SELECT PaymentMethod,
127             COUNT(*) AS TotalTransactions
128      FROM Orders
129      GROUP BY PaymentMethod
130      ORDER BY TotalTransactions DESC;
```

146 % No issues found Lr: 126, Ch: 1 (123 chars, 5 lines) TABS CRLF UTF-8

PaymentMethod	TotalTransactions
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

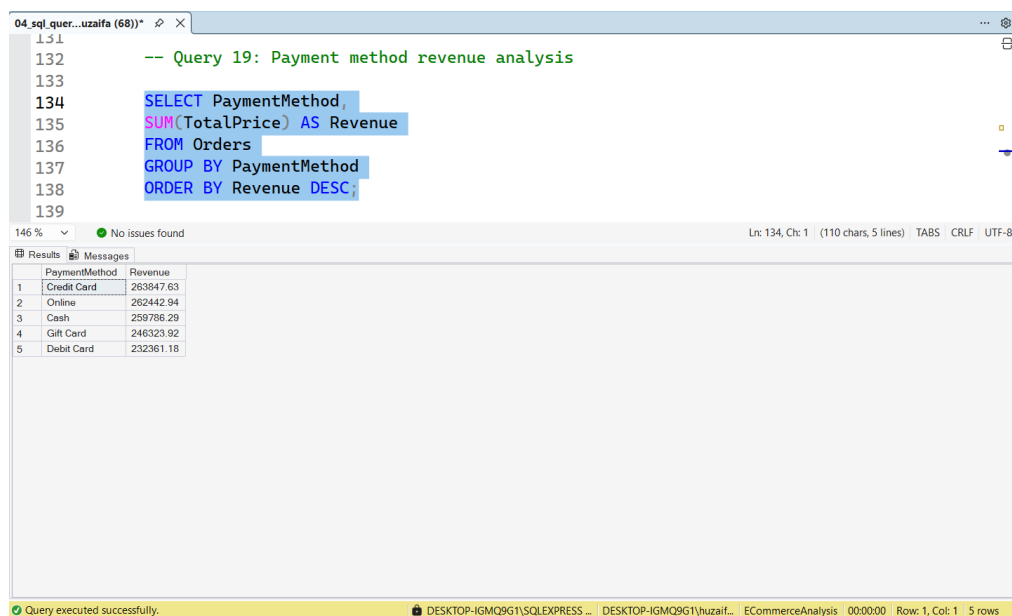
Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzalf... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 5 rows

Query 19. Payment Method Revenue Analysis

The query was used to calculate revenue generated through each payment method.

Analysis

The results revealed that **Credit Card** payments generated the highest revenue of **263,847.63**, followed closely by **Online Payments (262,442.94)** and **Cash Payments (259,786.29)**. **Debit Card** generated the lowest revenue at **232,361.18**. This analysis helps evaluate the financial contribution of different payment channels.



The screenshot shows a SQL query editor window with the following SQL code:

```
-- Query 19: Payment method revenue analysis

SELECT PaymentMethod,
SUM(TotalPrice) AS Revenue
FROM Orders
GROUP BY PaymentMethod
ORDER BY Revenue DESC;
```

The results of the query are displayed in a table with the following data:

PaymentMethod	Revenue
Credit Card	263847.63
Online	262442.94
Cash	259786.29
Gift Card	246323.92
Debit Card	232361.18

Query 20. Order Status Analysis

The query was used to count orders according to their current status.

Analysis

The results showed that **Cancelled Orders** were the highest with **250 orders**, followed by **Returned Orders (247)** and **Pending Orders (237)**. **Delivered Orders** accounted for **231 orders**. This analysis provides insights into order fulfillment performance and highlights areas where operational improvements may be required.

```
04_sql_quer...uzaifa (68)) *  X
139
140 -- Query 20: Order status analysis
141
142 SELECT OrderStatus,
143 COUNT(*) AS TotalOrders
144 FROM Orders
145 GROUP BY OrderStatus
146 ORDER BY TotalOrders DESC;
147
```

146 % No issues found Lnc 142, Ctc 1 (107 chars, 5 lines) TABS CRLF UTF-8

OrderStatus	TotalOrders
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzair... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 5 rows

Query 21. Referral Source Analysis

The query was used to count orders generated from different referral sources.

Analysis

The results indicated that **Instagram** generated the highest number of orders with **259 transactions**, followed by **Email (250 orders)** and **Google (241 orders)**. **Referral** generated the lowest number of orders with **222 transactions**. This analysis helps identify the most effective customer acquisition channels.

```
04_sql_quer...uzaifa (68)) *  X
147
148 -- Query 21: Referral source analysis
149
150 SELECT ReferralSource,
151 COUNT(*) AS TotalOrders
152 FROM Orders
153 GROUP BY ReferralSource
154 ORDER BY TotalOrders DESC;
155
```

146 % No issues found Lnc 150, Ctc 1 (113 chars, 5 lines) TABS CRLF UTF-8

ReferralSource	TotalOrders
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

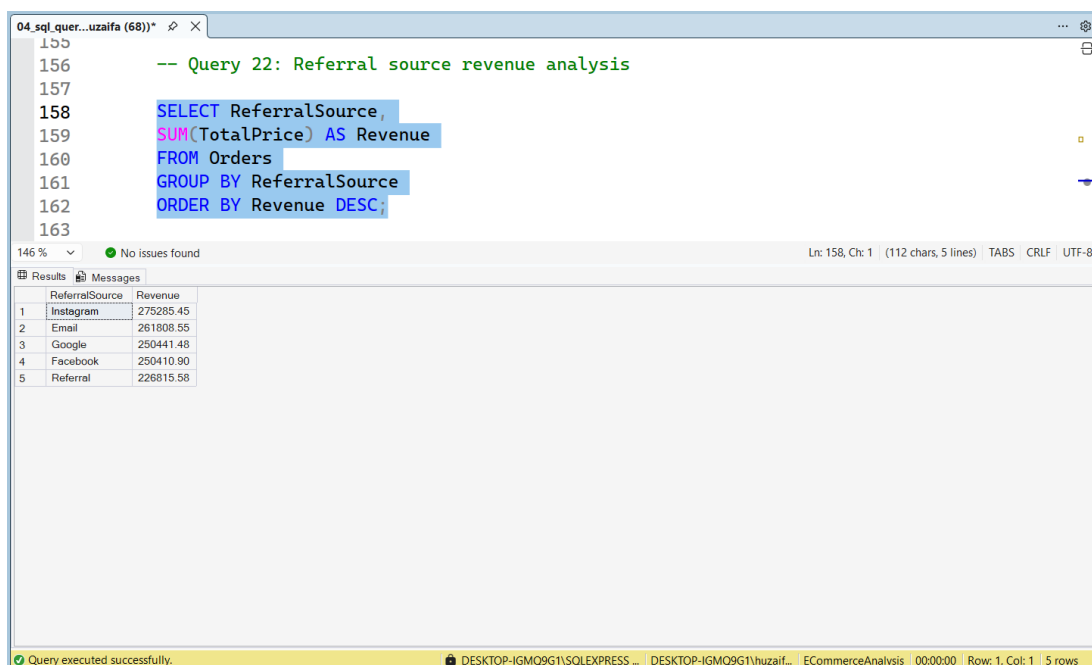
Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzair... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 5 rows

Query 22. Referral Source Revenue Analysis

The query was used to calculate revenue generated by each referral source.

Analysis

The results revealed that **Instagram** generated the highest revenue of **275,285.45**, followed by **Email (261,808.55)** and **Google (250,441.48)**. **Referral** generated the lowest revenue of **226,815.58**. These findings help evaluate the profitability of marketing channels and support future advertising strategies.



The screenshot shows a SQL query editor window titled "04_sql_quer...uzaifa (68)". The query is as follows:

```
-- Query 22: Referral source revenue analysis

SELECT ReferralSource,
SUM(TotalPrice) AS Revenue
FROM Orders
GROUP BY ReferralSource
ORDER BY Revenue DESC;
```

The results are displayed in a table with the following data:

ReferralSource	Revenue
Instagram	275285.45
Email	261808.55
Google	250441.48
Facebook	250410.90
Referral	226815.58

The status bar at the bottom indicates "Query executed successfully." and "DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzaiif... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 5 rows".

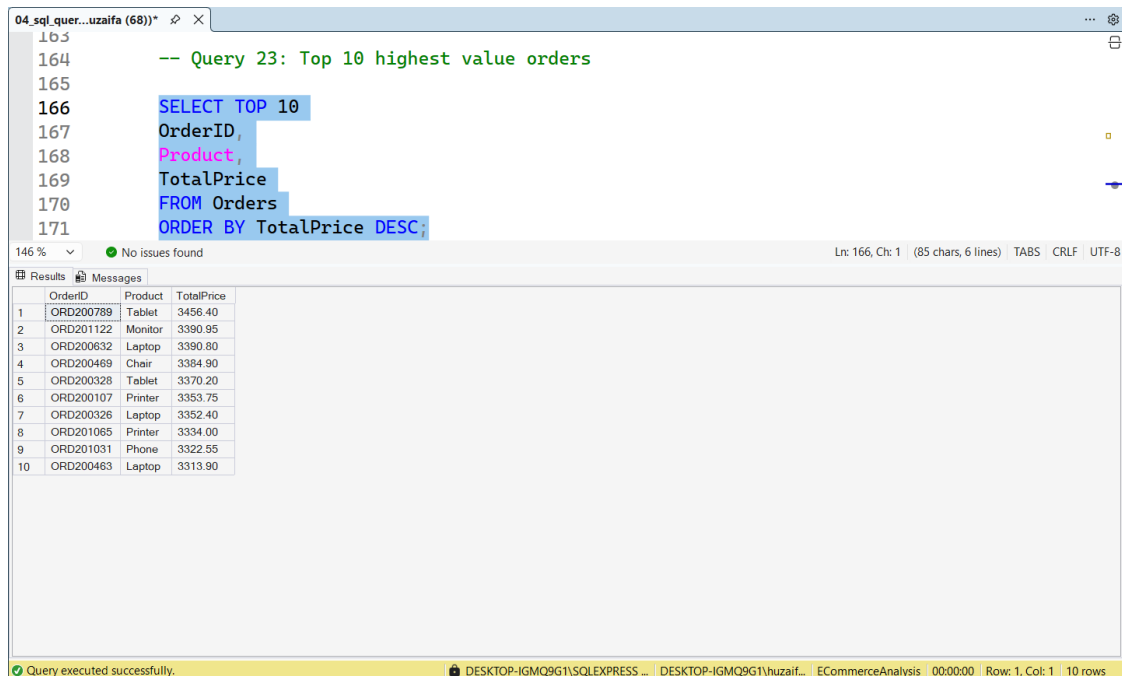
Query 23. Top 10 Highest Value Orders

The query was used to display the highest-value transactions in the dataset.

Analysis

The results identified the top ten customer orders based on total order value. The highest-value transaction was **Order ID ORD200789 (Tablet)** with a total value of **3456.40**. Other high-value transactions included **ORD201122 (Monitor - 3390.95)** and **ORD200632 (Laptop -**

3390.80). This analysis helps identify premium purchases and valuable customer transactions that contribute significantly to overall revenue.



The screenshot shows a SQL query editor window with a query titled "Query 23: Top 10 highest value orders". The query is as follows:

```
-- Query 23: Top 10 highest value orders
SELECT TOP 10
  OrderID,
  Product,
  TotalPrice
FROM Orders
ORDER BY TotalPrice DESC;
```

The results of the query are displayed in a table with 10 rows and 3 columns: OrderID, Product, and TotalPrice.

	OrderID	Product	TotalPrice
1	ORD200789	Tablet	3456.40
2	ORD201122	Monitor	3390.95
3	ORD200632	Laptop	3390.80
4	ORD200469	Chair	3384.90
5	ORD200328	Tablet	3370.20
6	ORD200107	Printer	3353.75
7	ORD200326	Laptop	3352.40
8	ORD201065	Printer	3334.00
9	ORD201031	Phone	3322.55
10	ORD200463	Laptop	3313.90

The status bar at the bottom indicates "Query executed successfully." and "Row: 1, Col: 1 | 10 rows".

Section 4. Advanced Analysis

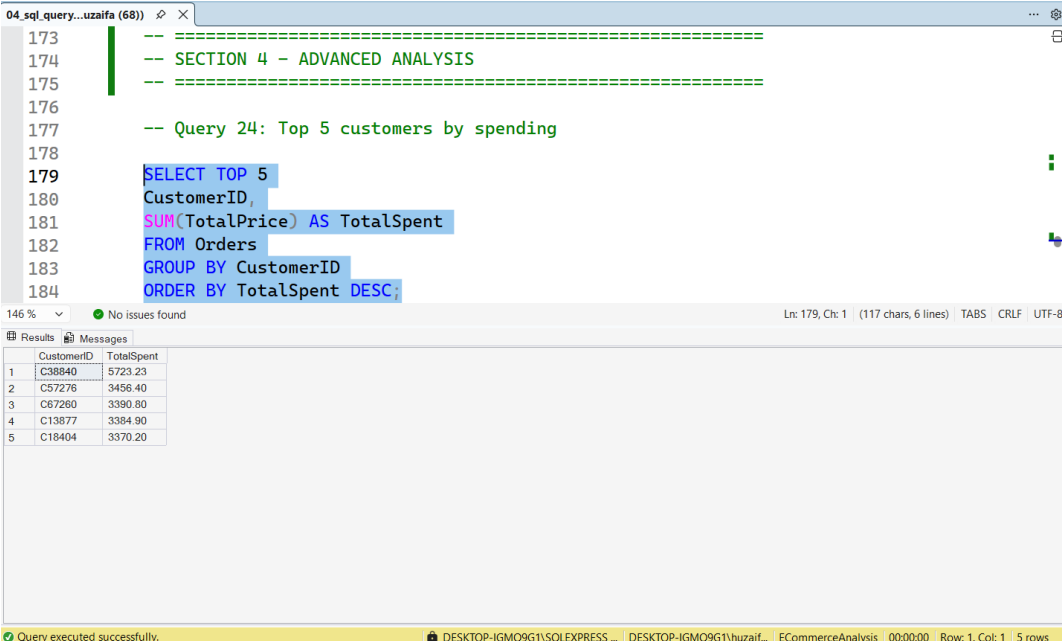
After performing business-oriented analysis, additional advanced SQL queries were executed to gain deeper insights into customer behavior, product pricing, and revenue trends. These queries demonstrate the use of advanced aggregation techniques and analytical reporting to support strategic business decision-making. The analysis focuses on identifying high-value customers, evaluating average product prices, measuring yearly business performance, and tracking monthly revenue patterns.

Query 24. Top 5 Customers by Spending

The query was used to identify the top five customers based on their total spending.

Analysis

The results revealed the highest-spending customers in the dataset. Customer **C38840** generated the highest total spending of **5,723.23**, followed by **C57276 (3,456.40)**, **C67260 (3,390.80)**, **C13877 (3,384.90)**, and **C18404 (3,370.20)**. Identifying these valuable customers helps businesses develop loyalty programs, improve customer retention, and target premium customers with personalized offers.



The screenshot shows a SQL query editor with a query window titled "04_sql_query...uzaifa (68)". The query is as follows:

```
-- =====  
-- SECTION 4 - ADVANCED ANALYSIS  
-- =====  
  
-- Query 24: Top 5 customers by spending  
  
SELECT TOP 5  
    CustomerID,  
    SUM(TotalPrice) AS TotalSpent  
FROM Orders  
GROUP BY CustomerID  
ORDER BY TotalSpent DESC;
```

The results pane shows the following data:

	CustomerID	TotalSpent
1	C38840	5723.23
2	C57276	3456.40
3	C67260	3390.80
4	C13877	3384.90
5	C18404	3370.20

The status bar at the bottom indicates "Query executed successfully." and "DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzai... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 5 rows".

Query 25. Average Product Price

The query was used to calculate the average unit price for each product category.

Analysis

The results showed that **Phone** products had the highest average unit price (**375.22**), followed by **Tablet (367.68)**, **Monitor (358.66)**, and **Laptop (357.71)**. The lowest average price was observed for **Desk (329.61)**. This analysis helps the business understand pricing differences among products and supports pricing strategy and profitability analysis.

```
04_sql_query...uzaifa (68)
185
186      -- Query 25: Average product price
187
188      SELECT Product,
189             AVG(UnitPrice) AS AverageUnitPrice
190      FROM Orders
191      GROUP BY Product
192      ORDER BY AverageUnitPrice DESC;
```

146 % No issues found Ln: 188, Cht: 1 (115 chars, 5 lines) TABS CRLF UTF-8

	Product	AverageUnitPrice
1	Phone	375.217564
2	Tablet	367.679106
3	Monitor	358.663619
4	Laptop	357.707052
5	Chair	355.660280
6	Printer	351.714917
7	Desk	329.608117

Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzalf... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 7 rows

Query 26. Revenue by Year

The query was used to calculate total revenue generated in each year.

Analysis

The results indicated that **2023** generated the highest revenue of **552,643.24**, followed by **2024** with **480,235.87**, while **2025** generated **231,882.85**. The analysis provides a clear overview of yearly business performance and helps management evaluate growth trends and financial outcomes over time.

```
04_sql_query...uzaifa (68)
193
194      -- Query 26: Revenue by year
195
196      SELECT YEAR(OrderDate) AS OrderYear,
197             SUM(TotalPrice) AS Revenue
198      FROM Orders
199      GROUP BY YEAR(OrderDate)
200      ORDER BY OrderYear;
```

146 % No issues found Ln: 196, Cht: 1 (124 chars, 5 lines) TABS CRLF UTF-8

	OrderYear	Revenue
1	2023	552643.24
2	2024	480235.87
3	2025	231882.85

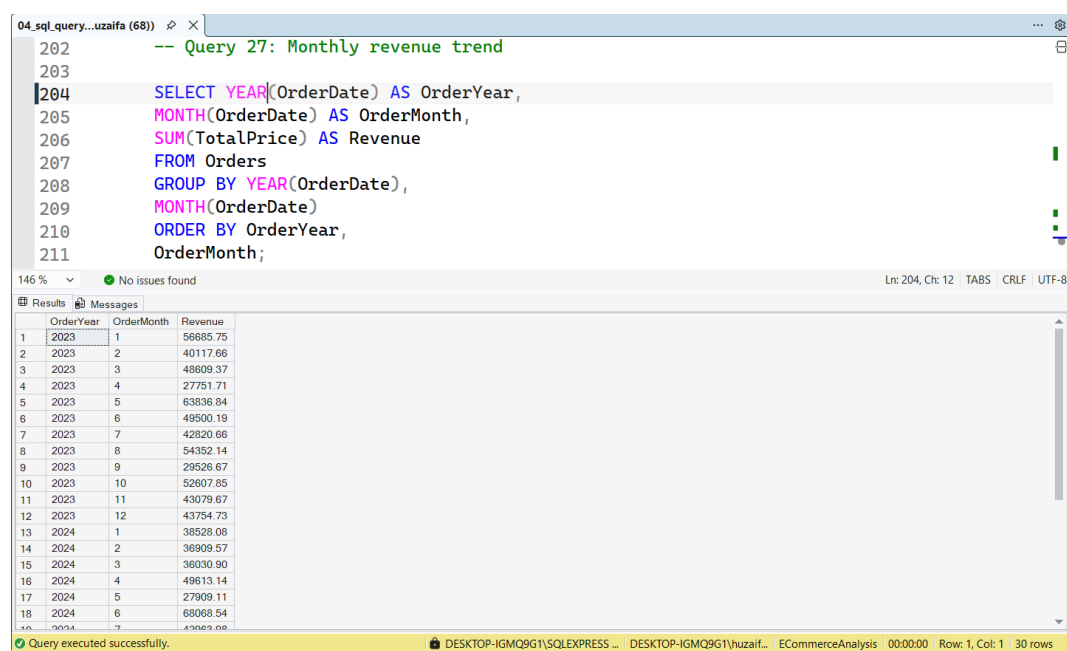
Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzalf... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 3 rows

Query 27. Monthly Revenue Trend

The query was used to calculate monthly revenue generated throughout the dataset period.

Analysis

The results provided a month-wise breakdown of revenue across multiple years. Revenue fluctuated throughout the reporting period, with some months generating significantly higher sales than others. For example, **May 2023 generated 63,836.84**, while **June 2024 generated 68,068.54**, representing some of the strongest revenue-performing months. This analysis helps identify seasonal sales patterns, forecast future demand, optimize inventory planning, and support strategic business decision-making.



The screenshot displays a SQL query window with the following text:

```
-- Query 27: Monthly revenue trend
202
203
204 SELECT YEAR(OrderDate) AS OrderYear,
205        MONTH(OrderDate) AS OrderMonth,
206        SUM(TotalPrice) AS Revenue
207 FROM Orders
208 GROUP BY YEAR(OrderDate),
209          MONTH(OrderDate)
210 ORDER BY OrderYear,
211          OrderMonth;
```

Below the query, the 'Results' tab shows a table with 30 rows. The first 12 rows represent the year 2023, and the next 18 rows represent the year 2024. The columns are OrderYear, OrderMonth, and Revenue.

OrderYear	OrderMonth	Revenue
2023	1	56685.75
2023	2	40117.66
2023	3	48609.37
2023	4	27751.71
2023	5	63836.84
2023	6	49500.19
2023	7	42820.66
2023	8	54352.14
2023	9	29526.67
2023	10	52807.85
2023	11	43079.67
2023	12	43754.73
2024	1	38528.08
2024	2	36909.57
2024	3	36030.90
2024	4	49613.14
2024	5	27909.11
2024	6	68068.54
2024	7	42069.08
2024	8	42069.08
2024	9	42069.08
2024	10	42069.08
2024	11	42069.08
2024	12	42069.08
2024	13	42069.08
2024	14	42069.08
2024	15	42069.08
2024	16	42069.08
2024	17	42069.08
2024	18	42069.08

The status bar at the bottom indicates: 'Query executed successfully. DESKTOP-IGMQ9G1\SQLEXPRESS ... DESKTOP-IGMQ9G1\huzait... ECommerceAnalysis 00:00:00 Row: 1, Col: 1 30 rows'.

Conclusion

This project demonstrated the practical application of SQL for data analysis using an e-commerce dataset. Various SQL techniques including data retrieval, filtering, sorting, grouping, aggregation, and business analysis were performed using Microsoft SQL Server.

The analysis provided valuable insights into customer purchasing behavior, product performance, revenue generation, payment preferences, referral source effectiveness, and sales trends. Key findings revealed that Chair was the top-selling product, Instagram generated the highest revenue among referral sources, Credit Card transactions contributed the highest payment-method revenue, and the year 2023 generated the highest overall revenue.

The project enhanced practical SQL skills and demonstrated how SQL can be used to transform raw business data into meaningful insights that support data-driven decision-making.